Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

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Abstract: **Perception models for autonomous vehicles are trained on large labeled datasets, and real driving footage records identifiable people, vehicles and places. Simulator-generated data is often treated as private, although simulators are calibrated with real statistics and trained models can reveal whether a record was part of their training set. This paper presents a pipeline that enforces differential privacy (DP) at the two points where real information enters. The Laplace mechanism privatizes seven aggregate statistics from the Waymo Open Dataset under a total budget of ε = 1.8 before they configure the CARLA simulator, and DP-SGD bounds the influence of each training record on the perception model. Every dataset file is encrypted with AES-256 and recorded in a SHA-256 manifest, labels are screened by a statistical poisoning detector, and a separate component verifies every signed claim before issuing a compliance certificate. On 10,000 synthetic frames from five towns, DP-SGD reduced the area under the curve (AUC) of a shadow-model membership inference attack from 0.639 to between 0.504 and 0.515 at ε = 1, 3 and 8, with a reduction of 0.16 to 0.21 in test AUC. On a deliberately overfitted model, the attack AUC fell from 0.628 to 0.501 at ε = 1 over five seeds ($p < 0.001$), which removed 99.5% of the attacker's advantage over random guessing, and smaller budgets gave significantly stronger protection. A YOLOv8n detector trained on the synthetic data achieved 0.366 mAP50 on synthetic images and retained 10.46% of the real-data mAP50 on KITTI, which identifies the domain gap between rendered and real images as the main limitation on utility.**

Keywords: differential privacy, synthetic data, autonomous vehicles, membership inference, DP-SGD, CARLA, data poisoning, GDPR

I. Introduction

Camera-based perception for autonomous vehicles is trained on large labeled datasets such as KITTI [1] and the Waymo Open Dataset [2]. Real recordings capture faces, license plates, locations and movement patterns, which are personal data under the GDPR [3]. Article 10 of the EU AI Act [4] further requires the training data of high-risk systems to be subject to data governance and to be relevant and, as far as possible, free of errors. Simulators such as CARLA [5] produce labeled images in any quantity without recording real people, and synthetic data has long been used to train perception models [6], [7].

Synthetic data is not private by construction. A simulator produces realistic scenes only when it is calibrated with real statistics such as traffic density, speed and weather, and this calibration carries information about the source data into the generated dataset. Stadler et al. [8] showed that synthetic data alone does not reliably protect source records against inference attacks. Trained models form a second channel, since membership inference attacks can determine whether a record was used for training [9], [10]. In addition, privacy claims are usually made by the developers of a system and are seldom verified independently.

Existing work typically protects a single stage, for example by training a generative model with DP [11], [12] or the downstream model with DP-SGD [13], and often reports a formal ε without an attack that tests the guarantee in practice. This work addresses the complete data path of a simulator-based driving dataset. The contributions are as follows:

- A pipeline that applies DP where real information enters: Laplace noise on seven calibration statistics with a total budget of ε = 1.8 under sequential composition, and DP-SGD with Rényi DP accounting and a fixed budget ceiling during training.
- Empirical validation with two membership inference attacks and a stress test on an overfitted model that serves as a positive control, reported over several seeds with bootstrap confidence intervals and paired significance tests.
- Security controls along the data path, comprising AES-256 encryption at capture, SHA-256 manifests, role-based access control, an audited authenticated encryption mode and a statistical label poisoning detector.
- An independent verification component that re-checks every signed claim with separate code and that detected a budget reporting error in the calibration release.
- A measurement of utility through cross-domain YOLOv8 evaluation on the KITTI benchmark.

Section II reviews related work, Section III describes the system and Section IV the experimental setup. Section V

reports the results, Section VI discusses them together with the limitations, and Section VII concludes the paper.

II. Related Work

A. Differential Privacy and DP-SGD

A randomized mechanism M is (ε, δ)-differentially private if, for any two datasets D and D′ that differ in one record and any set of outputs S, $\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S] + \delta$ [14], [15]. The Laplace mechanism satisfies pure ε-DP for numeric queries by adding noise scaled to the sensitivity of the query, and the budgets of mechanisms that read the same records add up under sequential composition [16]. DP-SGD [13] clips each per-sample gradient and adds Gaussian noise, and Rényi DP [17] provides tight accounting over many training steps. Opacus [18] implements both methods for PyTorch.

B. Membership Inference and Privacy Auditing

Shokri et al. [9] introduced shadow-model membership inference, in which an attack model learns to separate members from non-members using the outputs of the target model. Yeom et al. [10] related this risk to overfitting and proposed a loss-threshold attack. Carlini et al. [19] argued that attacks should be evaluated by their true positive rate at a low false positive rate in addition to AUC. Jagielski et al. [20] used attacks to audit DP-SGD and found that measured leakage is often far below the formal bound. The present work follows this auditing approach and reports both the formal ε and the measured attack performance.

C. Synthetic Data for Driving

Game engines and simulators provide pixel-accurate labels at low cost [6], [7], [5]. The principal obstacle is the domain gap between rendered and real images, which domain randomization [21] and mixed training reduce but do not eliminate. Utility is commonly measured by training on synthetic data and testing on a real benchmark such as KITTI [1], and the distance between image distributions by the Fréchet Inception Distance (FID) [22].

D. Privacy of Synthetic Data, Poisoning and Compliance

DP generative models such as PATE-GAN [11] and DPGAN [12] protect the records used to train the generator. A simulator pipeline has no trained generator, and its privacy-relevant input is the set of calibration statistics; these statistics are therefore the target of privatization in this work. Label poisoning can degrade a model without any change to the images [23], [24], so labels require a separate check. For released metadata, k-anonymity [25] measures how easily a record can be singled out through quasi-identifiers.

III. System Design

The system consists of four components, each developed by one author and connected only through files with fixed formats (Fig. 1). Privacy Integration privatizes the calibration statistics and trains and attacks the models. Dataset Generation captures and protects the data. Utility Evaluation verifies the files, screens the labels and measures detection utility. Risk and Compliance verifies the evidence produced by the other three components.

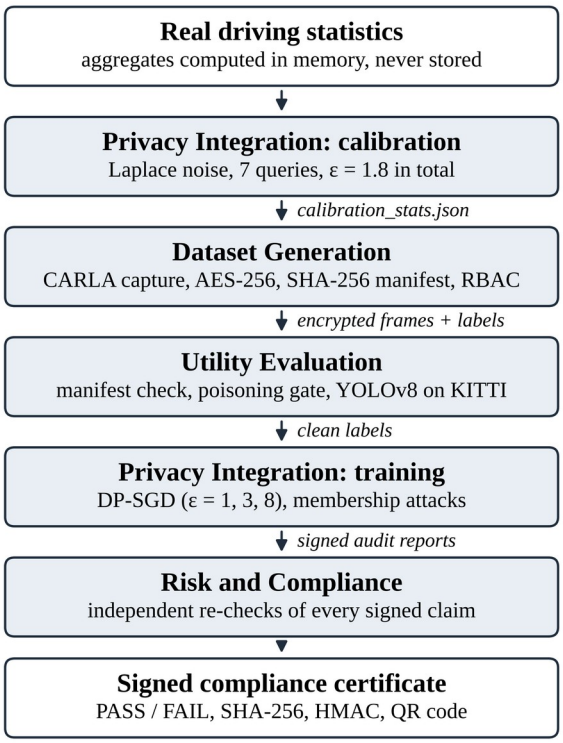


Fig. 1. Pipeline overview. Each box is a separate component, and arrows show the files passed between components.

The threat model includes four adversaries: an adversary who obtains the released calibration statistics and attempts to recover the source values; an adversary with query access to a trained model who attempts to determine whether a frame was in its training set; an adversary who can read or modify dataset files, including label files, in storage or in transit; and an insider who alters evidence such as a privacy report after it has been produced. Compromise of the key store and side channels on the training machine are outside the scope of this work.

A. Privacy-Aware Calibration

Seven aggregate statistics configure the simulator: vehicle density, pedestrian density, average vehicle speed, average pedestrian speed and intersection count per kilometer, together with the weather and time-of-day distributions. They are computed in memory from the Waymo Open Dataset [2], and no raw record is written to disk. Each statistic f is released through the Laplace mechanism

$$M(D) = f(D) + \mathrm{Lap}(\Delta f / \varepsilon_i), \tag{1}$$

where Δf is the sensitivity of the query (1 for the scalar statistics, except average vehicle speed, for which it is 2) and ε_i is the budget of query i. Because all seven queries read the same records, their budgets compose sequentially:

$$\varepsilon_{\text{total}} = \Sigma_i \, \varepsilon_i = 7 \times 0.2571 = 1.8. \tag{2}$$

Each query is also capped at ε = 2. Only the privatized file leaves the component; the true values and the noise drawn for each query are written to an audit log that is shared only with Risk and Compliance. A noised value may fall outside its valid range, as the published intersection count does. Mapping such a value into range when configuring the simulator is post-processing and consumes no additional budget [15].

B. Secure Dataset Generation

CARLA runs in synchronous mode at 20 frames per second, and one frame set is saved every 16 ticks (0.8 s) to avoid near-duplicate consecutive samples. Five towns (Town01 to Town05) contribute 2,000 sets each. A set contains an RGB image, a depth map and a semantic segmentation image at 800 × 600 pixels, a class mask and a YOLO label file for three classes: car, pedestrian and cyclist. Vehicle and pedestrian counts, speeds, weather and time of day are taken from the privatized statistics, and a configuration record for each town stores the profile used, so that the link can be audited.

Each image is hashed with SHA-256 and encrypted with AES-256 [26], and the plaintext is deleted only after the ciphertext has been written completely. The delivered dataset uses CBC mode with a random 16-byte initialization vector. The key is read from an environment file excluded from version control, and the capture stops if the key is missing instead of generating a random key that no other component could use. A role-based gateway restricts the creation, encryption and verification of data. The encryption engine also supports GCM [27], whose authentication tag causes a modified file to fail decryption. A security audit script in the same component tests either mode and scans captured folders.

C. Utility Evaluation and Poisoning Detection

Utility Evaluation decrypts each file, recomputes its SHA-256 digest and compares it with the manifest, so that a file modified in transit is rejected. The labels then pass a three-layer poisoning detector. The structural layer rejects malformed lines, unknown class identifiers, coordinates outside [0, 1], degenerate boxes and implausible object counts. The statistical layer compares each batch of 100 frames with the remainder of its town using a chi-square goodness-of-fit test and rejects the batch only if the deviation is both significant ($p < 0.0001$) and large (Cramér's $V > 0.40$). The cross-town layer compares the class shares of each town with those of the other four and issues a warning only, since towns differ by design. As a positive control, an out-of-range box, an unknown class and a fully relabeled batch are injected into a scratch copy, and all three must be detected.

Utility is measured with YOLOv8n [28] in four experiments: training and testing on synthetic data, training on synthetic and testing on real data, training and testing on real data, and training on real and testing on synthetic data. KITTI [1] supplies the real data. It is independent of the calibration source, which avoids a circular evaluation. Utility retention is defined as the synthetic-to-real score expressed as a percentage of the real-to-real score.

D. Private Training and Attack Validation

The private model is a convolutional classifier that predicts whether a frame contains a pedestrian or a cyclist. It operates on 448 × 336 inputs, which preserve the 4:3 aspect ratio, and uses group normalization because Opacus does not support batch normalization. DP-SGD [13] computes the noisy gradient

$$\tilde{g} = (1/B)\,[\Sigma_i \,\mathrm{clip}(g_i, C) + N(0, \sigma^2 C^2 I)], \qquad (3)$$

with clipping norm $C = 1.0$, $\delta = 10^{-5}$, Poisson sampling and Rényi DP accounting [17], [18]. Models are trained at target budgets of $\varepsilon = 1$, 3 and 8, which require noise multipliers $\sigma = 0.977$, 0.665 and 0.507, together with a non-private baseline of the same architecture. DP training uses SGD with momentum, because the second-moment estimate of Adam accumulated the injected noise and destabilized training. A ceiling of $\varepsilon = 10$ halts any run that would exceed it.

Two attacks are applied to each model. The shadow-model attack [9] uses two features: the confidence on the true label and the per-sample loss. Member and non-member evaluation sets have equal size, so a success rate of 50% corresponds to random guessing. The loss-threshold attack [10] requires no shadow model and is reported as the true positive rate (TPR) at false positive rates (FPR) of 1% and 5% [19].

A reduction in attack performance can only be observed if the baseline model leaks. A stress test was therefore added in which every target is trained on 500 members for 60 epochs, so that the non-private model overfits. The DP models are trained under the same conditions and evaluated with the same loss-threshold attack, which attributes any drop in attack AUC to DP. Each run reports a 95% confidence interval from 1,000 bootstrap resamples, and paired one-sided t-tests over seeds test the reduction for each budget and the ordering between adjacent budgets.

Each training run writes a per-step budget log and an audit report signed with HMAC-SHA256 [29]. The signing key is stored separately from the dataset encryption key.

E. Independent Risk and Compliance Verification

Risk and Compliance is a separate code base that verifies every input it receives. It performs seven checks: (i) the signature and digest of every report; (ii) a budget audit that compares each reported ε with the per-step log and requires the logged values to be non-decreasing and below the ceiling; (iii) a reconciliation of the calibration audit log, in which the true value plus the logged noise must equal the published value; (iv) a composition check of the calibration budget; (v) k-anonymity [25] of the released metadata over weather, time of day, event type and town, with $k \geq 5$; (vi) re-verification of every manifest hash; and (vii) acceptance of the poisoning detector verdict. The component also runs a reconstruction attack on the calibration channel and a pattern-based scan of the metadata for file paths, e-mail addresses, IP addresses, GPS coordinates, credentials and machine identifiers. The results are mapped to eight GDPR controls under Articles 5, 17, 25 and 32 and summarized in a certificate that contains the SHA-256 digest of the report, an HMAC signature and a QR code. The certificate records the outcome of the listed checks and is not a legal certification of compliance.

IV. Experimental Setup

The synthetic dataset comprises 10,000 annotated frames from five towns with 15,084 objects: 11,008 cars, 2,424 pedestrians and 1,652 cyclists. Of these frames, 32.6% contain no object, and 86.4% of pedestrians are smaller than 32 × 32 pixels. The data were split 70/15/15 into training, validation and test sets with seed 42 (7,000, 1,500 and 1,500 frames). The KITTI training set (7,481 images) was mapped to the same three classes and split in the same way (5,236, 1,122 and 1,123 images).

Both detectors are YOLOv8n models trained for up to 50 epochs at 640 × 640 pixels with a batch size of 8 and early stopping after 15 epochs without improvement. All scores are measured on the test splits, which are not used for early stopping. FID is computed on 1,000 synthetic test images, approximately 200 per town, and 1,000 KITTI test images.

In the DP study, each configuration is trained with seeds 42, 43 and 44 for 10 epochs with a batch size of 32, and the membership evaluation sets contain 1,179 members and 1,179

non-members. The stress test uses seeds 42 to 46. Python, NumPy and PyTorch are seeded and cuDNN autotuning is disabled; repeating a run with a fixed seed reproduced the same per-epoch loss and the same test AUC.

An earlier version of the calibration module ran each of the seven queries at ε = 1.8 and therefore spent a total of 12.6 while reporting 1.8. The composition check described in Section III-E detected this error. The corrected module divides the total of 1.8 across the seven queries. The dataset was captured again from the corrected release, and all results in Section V were obtained on this capture.

V. Results

A. Calibration Privacy

The reconstruction attack compared the five published scalar statistics with the true values. At ε = 0.2571 per query, the mean absolute error was 4.37, against an expected Laplace error of 4.67, a ratio of 0.94. The noise therefore has the magnitude required by the mechanism, and the source values cannot be recovered from the release. The published weather distribution was 0.650 clear, 0.156 cloudy, 0.195 rain and 0 fog. The composition check passes at 1.8 on the corrected release and fails at 12.6 on the earlier one.

B. Membership Inference with DP-SGD

TABLE I. Privacy and utility of the trained models (mean ± SD over three seeds)

Model	Achieved ε	Test AUC	Attack AUC
No DP	none	0.829 ± 0.009	0.639 ± 0.013
ε = 1	0.991	0.621 ± 0.045	0.504 ± 0.002
ε = 3	3.000	0.655 ± 0.035	0.515 ± 0.002
ε = 8	7.995	0.668 ± 0.020	0.506 ± 0.007

Attack AUC of the shadow-model attack; 0.5 corresponds to random guessing.

Table I summarizes the main study. Against the non-private model, the shadow-model attack reached an AUC of 0.639 and a success rate of 56.7%. Against the DP models, the AUC fell to between 0.504 and 0.515 and the success rate to between 50.3% and 51.6%. Over the nine DP runs the attack AUC ranged from 0.499 to 0.516 and did not overlap the baseline range of 0.625 to 0.650. Paired one-sided t-tests over seeds gave p = 0.0013, 0.0016 and 0.0015 for ε = 1, 3 and 8. The loss-threshold attack gave consistent results, with an AUC of 0.610 without DP and 0.501 to 0.506 with DP. The utility cost was significant: test AUC decreased by 0.21, 0.17 and 0.16 for ε = 1, 3 and 8 (p = 0.0099, 0.0084 and 0.0031). Every run ended at its target budget, well below the ceiling.

The leakage of the baseline in this setting is modest and average-case. At 1% FPR, the loss-threshold attack on the non-private model identified 0.7% of members, which is no better than chance. Since all three budgets lie close to chance, the main study cannot distinguish between them.

C. Stress Test

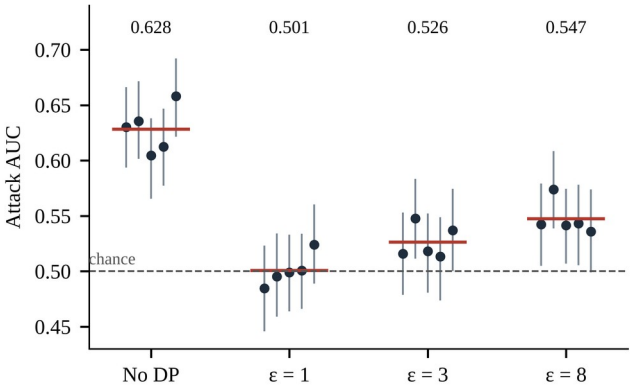


Fig. 2. Stress test: attack AUC per seed with 95% bootstrap intervals. Red lines mark the means over five seeds.

TABLE II. Stress test on an overfitted model (five seeds)

Model	Attack AUC	p (one-sided)	Advantage removed
No DP	0.628		
ε = 1	0.501	0.00004	99.5%
ε = 3	0.526	0.00006	79.4%
ε = 8	0.547	0.001	63.0%

Advantage removed: reduction in attack AUC as a share of the baseline advantage over 0.5.

In the stress test the non-private model leaked clearly (Fig. 2, Table II). Its attack AUC averaged 0.628, and the 95% interval of every seed lay entirely above 0.5, so the positive control passed. DP reduced the attack at every budget. At ε = 1 the AUC fell to 0.501, which removed 99.5% of the attacker's advantage over random guessing. The budgets were also ordered as theory predicts: ε = 1 gave stronger protection than ε = 3 (p = 0.013), and ε = 3 stronger protection than ε = 8 (p = 0.010).

D. Utility

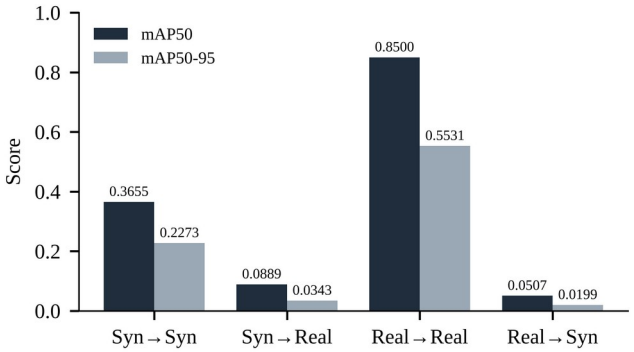


Fig. 3. Cross-domain detection with YOLOv8n (training domain → test domain).

TABLE III. Detection results per experiment (test splits)

Train → test	mAP50	mAP50-95	Car	Pedestrian
Syn → Syn	0.3655	0.2273	0.608	0.179
Syn → Real	0.0889	0.0343	0.261	0.005
Real → Real	0.8500	0.5531	0.952	0.777
Real → Syn	0.0507	0.0199	0.127	0.008

Car and Pedestrian columns give per-class AP50. Syn: synthetic; Real: KITTI.

The detector trained on synthetic data reached 0.366 mAP50 on synthetic test images (Fig. 3, Table III). On KITTI it reached 0.089, which is 10.46% of the 0.850 obtained by training on KITTI, or 6.2% under the stricter mAP50-95 metric. Cars transferred best (0.261 AP50), whereas pedestrians, most of which are smaller than 32 × 32 pixels, reached only 0.005. The gap was also present in the reverse

direction, where the KITTI model scored 0.051 on synthetic images. The FID between the two test sets was 115.99.

E. Poisoning Detection and Encryption

On the delivered labels, the structural layer found no anomaly among 15,084 boxes. The statistical layer rejected 24 of 100 batches, which removed 2,400 frames, and the cross-town layer reported a cyclist share of 0.030 in Town01 against 0.116 in the other towns (9.8 standard deviations). No poisoning was injected into these labels, so the rejected batches reflect natural variation in scene content within a town, such as consecutive frames along one route. The positive control detected all three injected defects. The detector results reported above use the 70/15/15 split of all 10,000 frames, without this filtering.

An audit of the GCM mode passed all five synthetic tests. Decryption restored the original bytes; 30 encryptions of the same file produced 30 distinct nonces; the chi-square statistic of the ciphertext bytes was 272.59, below the critical value of 293.25 for 255 degrees of freedom; a one-bit change in the input altered 50.0% of the ciphertext bits; and a file with one modified byte was rejected. In a capture of 6,093 Town01 images, none of the encrypted files could be read as an image without the key. GCM adds 32 bytes to each file.

F. Independent Verification

TABLE IV. INDEPENDENT CHECKS BY RISK AND COMPLIANCE

Check	Result	Verdict
Report signatures	12 of 12 signed reports verified	PASS
Budget audit	Reported ε equals the log after 3,090 steps	PASS
Calibration log	True value + noise = published value	PASS
Composition	7 × 0.2571 = 1.8	PASS
k-anonymity	k = 5 over 10,000 rows (192 classes)	PASS
Manifests	40,000 file hashes match	PASS
Poisoning detection	Verdict FILTERED accepted	PASS

Also performed: reconstruction attack (error ratio 0.94) and leakage scan (0 findings in 9 files).

All seven checks in Table IV passed on the final outputs. The budget audit matched the reported budgets to the final logged values of 0.990572, 2.99984 and 7.995129. All eight GDPR controls were verified, and the certificate verdict was PASS. Earlier in the project, the composition check detected the 12.6 error described in Section IV, which had not been noticed by the component that produced the statistics.

VI. DISCUSSION

DP-SGD reduced the attack to chance at every tested budget, including ε = 8, where the formal bound is loose. This agrees with the finding that practical attacks often fall well short of the worst case [20]. The stress test adds two results that the main study cannot provide. It confirms that the attacks are effective, since they succeed against a model that leaks, and it shows that the budgets behave as expected once there is leakage to remove.

The utility results separate two costs. At the model level, DP-SGD reduced test AUC by 0.16 to 0.21. At the data level, utility is limited mainly by the domain gap. Calibration noise acts on scene parameters and leaves the rendering unchanged, and the gap is large in both directions, which suggests that rendering, image shape (4:3 compared with the wide KITTI format) and small object sizes account for most of it. Because no non-private reference dataset was captured, the share of the gap caused by calibration noise cannot be isolated, and it is not assumed to be zero.

The budget reporting error shows the value of independent verification. The error was present in the code of the component that produced the statistics and was found by a separate component that applied one rule: the per-query budgets must compose to the claimed total.

The work has the following limitations, all of which are also declared in the signed outputs:

- Opacus secure mode is disabled, so DP noise is drawn from a standard pseudo-random generator, because the library it requires does not support current PyTorch versions.
- Minority frames were oversampled by duplicating indices. A duplicated record contributes to more than one gradient, which weakens its per-record guarantee.
- Both attacks are average-case. A reference-calibrated attack following [19] has been implemented, but its results are not reported here.
- The poisoning detector rejected 24% of clean frames, which is a high false-positive rate.
- The delivered dataset uses CBC, which depends on the manifest rather than an authentication tag to detect modification, and deleted plaintext is not cryptographically erased.

VII. CONCLUSION AND FUTURE WORK

This paper presented a synthetic data pipeline for autonomous vehicle perception that enforces differential privacy at calibration and at training, protects every file, screens labels for poisoning and has every claim verified by an independent component. DP-SGD reduced membership inference to chance in the main study and removed 99.5% of the attacker's advantage on an overfitted model at ε = 1, with significant differences between budgets. The synthetic data supports learning in simulation, but transfer to real images is limited to 10.46% of the real-data mAP50.

Future work will reduce the domain gap through domain adaptation, mixed real and synthetic training and higher resolution; enable a secure random number generator for DP noise; report calibrated attacks with more reference models; adopt GCM across the pipeline; and use per-batch keys so that deleting a key provably erases the corresponding data.

ACKNOWLEDGMENT

The authors thank the Faculty of Computing of the Sri Lanka Institute of Information Technology for its support throughout this research. The authors are also grateful to the members of the evaluation panel for their feedback during the progress presentations, which led to the stress test and the positive controls reported in this paper. The authors acknowledge Waymo for providing the Waymo Open Dataset, the providers of the KITTI Vision Benchmark Suite, and the developers of the CARLA simulator and the Opacus library, whose openly available resources made this work possible.

REFERENCES

[1] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2012, pp. 3354-3361.

[2] P. Sun et al., "Scalability in perception for autonomous driving: Waymo Open Dataset," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2020, pp. 2446-2454.

[3] European Parliament and Council of the European Union, "Regulation (EU) 2016/679 (General Data Protection Regulation)," Off. J. Eur. Union, vol. L 119, pp. 1-88, May 2016.

[4] European Parliament and Council of the European Union, “Regulation (EU) 2024/1689 (Artificial Intelligence Act),” Off. J. Eur. Union, Jul. 2024.

[5] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, “CARLA: An open urban driving simulator,” in Proc. 1st Annu. Conf. Robot Learning (CoRL), 2017, pp. 1-16.

[6] S. R. Richter, V. Vineet, S. Roth, and V. Koltun, “Playing for data: Ground truth from computer games,” in Proc. Eur. Conf. Comput. Vis. (ECCV), 2016, pp. 102-118.

[7] G. Ros, L. Sellart, J. Materzynska, D. Vazquez, and A. M. Lopez, “The SYNTHIA dataset: A large collection of synthetic images for semantic segmentation of urban scenes,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 3234-3243.

[8] T. Stadler, B. Oprisanu, and C. Troncoso, “Synthetic data: Anonymisation groundhog day,” in Proc. 31st USENIX Security Symp., 2022, pp. 1451-1468.

[9] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, “Membership inference attacks against machine learning models,” in Proc. IEEE Symp. Security and Privacy (SP), 2017, pp. 3-18.

[10] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, “Privacy risk in machine learning: Analyzing the connection to overfitting,” in Proc. IEEE 31st Computer Security Foundations Symp. (CSF), 2018, pp. 268-282.

[11] J. Jordon, J. Yoon, and M. van der Schaar, “PATE-GAN: Generating synthetic data with differential privacy guarantees,” in Proc. Int. Conf. Learn. Represent. (ICLR), 2019.

[12] L. Xie, K. Lin, S. Wang, F. Wang, and J. Zhou, “Differentially private generative adversarial network,” arXiv:1802.06739, 2018.

[13] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, “Deep learning with differential privacy,” in Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS), 2016, pp. 308-318.

[14] C. Dwork, F. McSherry, K. Nissim, and A. Smith, “Calibrating noise to sensitivity in private data analysis,” in Proc. Theory of Cryptography Conf. (TCC), 2006, pp. 265-284.

[15] C. Dwork and A. Roth, “The algorithmic foundations of differential privacy,” Found. Trends Theor. Comput. Sci., vol. 9, no. 3-4, pp. 211-407, 2014.

[16] F. McSherry, “Privacy integrated queries: An extensible platform for privacy-preserving data analysis,” in Proc. ACM SIGMOD Int. Conf. Management of Data, 2009, pp. 19-30.

[17] I. Mironov, “Rényi differential privacy,” in Proc. IEEE 30th Computer Security Foundations Symp. (CSF), 2017, pp. 263-275.

[18] A. Yousefpour et al., “Opacus: User-friendly differential privacy library in PyTorch,” arXiv:2109.12298, 2021.

[19] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, “Membership inference attacks from first principles,” in Proc. IEEE Symp. Security and Privacy (SP), 2022, pp. 1897-1914.

[20] M. Jagielski, J. Ullman, and A. Oprea, “Auditing differentially private machine learning: How private is private SGD?” in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020, pp. 22205-22216.

[21] J. Tremblay et al., “Training deep networks with synthetic data: Bridging the reality gap by domain randomization,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW), 2018, pp. 969-977.

[22] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, “GANs trained by a two time-scale update rule converge to a local Nash equilibrium,” in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017, pp. 6626-6637.

[23] B. Biggio, B. Nelson, and P. Laskov, “Poisoning attacks against support vector machines,” in Proc. 29th Int. Conf. Mach. Learn. (ICML), 2012.

[24] J. Steinhardt, P. W. Koh, and P. Liang, “Certified defenses for data poisoning attacks,” in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017.

[25] L. Sweeney, “k-anonymity: A model for protecting privacy,” Int. J. Uncertainty, Fuzziness and Knowledge-Based Syst., vol. 10, no. 5, pp. 557-570, 2002.

[26] National Institute of Standards and Technology, “Advanced Encryption Standard (AES),” FIPS PUB 197, 2001.

[27] M. Dworkin, “Recommendation for block cipher modes of operation: Galois/Counter Mode (GCM) and GMAC,” NIST Special Publication 800-38D, 2007.

[28] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[29] H. Krawczyk, M. Bellare, and R. Canetti, “HMAC: Keyed-hashing for message authentication,” RFC 2104, 1997.